Indian Agriculture Crop Trend Analysis

Description

This project analyzes agricultural trends in India using crop-related data. It focuses on comparing crop production, cultivated area, yield, and available agricultural factors such as rainfall, temperature, fertilizer, and pesticide usage.

Microsoft Excel was used for data cleaning and preparation, and Power BI was used to build interactive dashboards and communicate findings through visualizations.

The aim is to explore patterns across crops, states, years, and seasons, and to present the results in a clear, easy-to-understand format.

Getting Started

Dependencies

- Microsoft Excel
- Power BI Desktop
- Dataset containing agricultural information
- Power Query and DAX, where applicable

Installing

1. Download or clone this project repository from GitHub.
2. Open the dataset in Microsoft Excel.
3. Review and clean the data before analysis.
4. Open Power BI Desktop.
5. Import the cleaned dataset into Power BI.
6. Save the Power BI report (`.pbix`) in the project folder.

Executing the Project

1. Understand the dataset

Review the dataset structure and the available columns. The project dataset has included fields such as:

- `Crop`
- `Year`
- `Season`
- `State`
- `Area`

- `Production`
- `Yield`
- `Fertilizer`
- `Pesticide`
- `Temperature`
- `Rainfall`

2. Perform data cleaning in Excel

Check and handle the following where necessary:

- Missing values
- Duplicate records
- Incorrect data types
- Inconsistent crop and state names
- Invalid or unexpected numeric values
- Inconsistent category formats

Keep a record of important cleaning decisions so the process can be explained and reproduced.

3. Prepare the data for Power BI

1. Save the cleaned dataset in a suitable format, such as CSV or Excel.
2. Open Power BI Desktop.
3. Select **Get Data** and import the cleaned file.
4. Check column data types and relationships, if applicable.
5. Create measures and calculated columns only where needed.

4. Create visualizations

Use suitable visuals to explore the data, such as:

- Crop-wise production comparison
- State-wise production comparison
- Yearly production trends
- Area and yield analysis
- Seasonal crop comparisons
- Rainfall and temperature trends
- Fertilizer and pesticide comparisons

Only include charts and measures that are present in the completed report.

5. Build an interactive dashboard

The dashboard include:

- KPI cards for key agricultural measures
- Bar charts for crop and state comparisons
- Line charts for yearly trends
- Comparison visuals for agricultural factors
- Slicers for year, state, crop, and season, where available

6. Identify insights

Use the visuals to answer questions such as:

- Which crops have the highest recorded production?
- How does production change over time?
- How do production and yield vary by state and season?
- What patterns appear between rainfall, temperature, and crop outcomes?
- How do fertilizer and pesticide usage vary in the dataset?

Report conclusions only after confirming them from the actual analysis. Relationships in the data do not automatically prove cause and effect.

Help

- If Power BI cannot locate the dataset, confirm that the file is saved in the expected folder.
- If values look incorrect, check the column data types and cleaning steps.
- Verify the units and definitions from the original dataset documentation.
- Use the cleaned dataset for dashboard creation.
- If a visual shows unexpected results, review filters, relationships, and measures.

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Version History

0.2

- Added data cleaning and preparation in Excel.
- Developed agricultural trend analysis visuals in Power BI.
- Added crop, state, year, and season comparison ideas.
- Explored production, area, yield, and available agricultural factors.

0.1

- Initial project setup.
- Identified agricultural dataset fields and analysis goals.

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- The original dataset provider and contributors.
- Microsoft Excel for data preparation.
- Microsoft Power BI for dashboard development.
- Learning resources and the data analytics community.